CSE558 · DATA SCIENCE — LECTURE 6

Closing the z-test, Opening the t-test

Today, in order

z-test for a proportion, then the t-test, χ^2-test

Not today

The two-sample family, χ^2 independence

After this

Straight into the class test

CLOSING THE GAP FROM LECTURE 5

z-test for a proportion

Use it when

- A sample proportion against a claimed population rate.
- n large — roughly 30 or more.
- H_0: $\mu = p_0$ · H_1: $\mu \neq p_0$ · $\alpha = 0.05$

$$Z = \frac{p - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

Same test, different version

A proportion is the mean of a string of 0/1 outcomes, and $\sqrt{(p_0(1-p_0)/n)}$ is exactly the standard error of that mean.

QUESTION

A quick check

A shop claims at least 80% of its deliveries arrive on time. You sample 50 deliveries and 38 are on time.

Is this a one-sided or two-sided test, and which side?

H_0 $\mu \geq 80\%$

H_1 $\mu < 80\%$.

Answer

H_0: $p \geq 0.80$ — the shop's claim, given the benefit of the doubt.
H_1: $p < 0.80$ — one-sided, left tail.

$\hat{p} = 38/50 = 0.76$. This is the direction that could reject H_0.

THE QUESTION EVERY Z-TEST HAS BEEN DODGING

When have you ever actually known σ?

Both z-tests so far assumed the population standard deviation

Not your sample's spread!!

What you actually have

n observations, their mean $\bar{x}$, and their spread s. That is the entire evidence base.

Today's fix

Substitute s for σ — and accept that the substitution costs something.

THE FIX

t-test for a population mean

Use it when

- One variable — a sample mean against a claimed population mean.
- σ is unknown, so you estimate it with $s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$.
- Works even when n is small — that is the whole point.

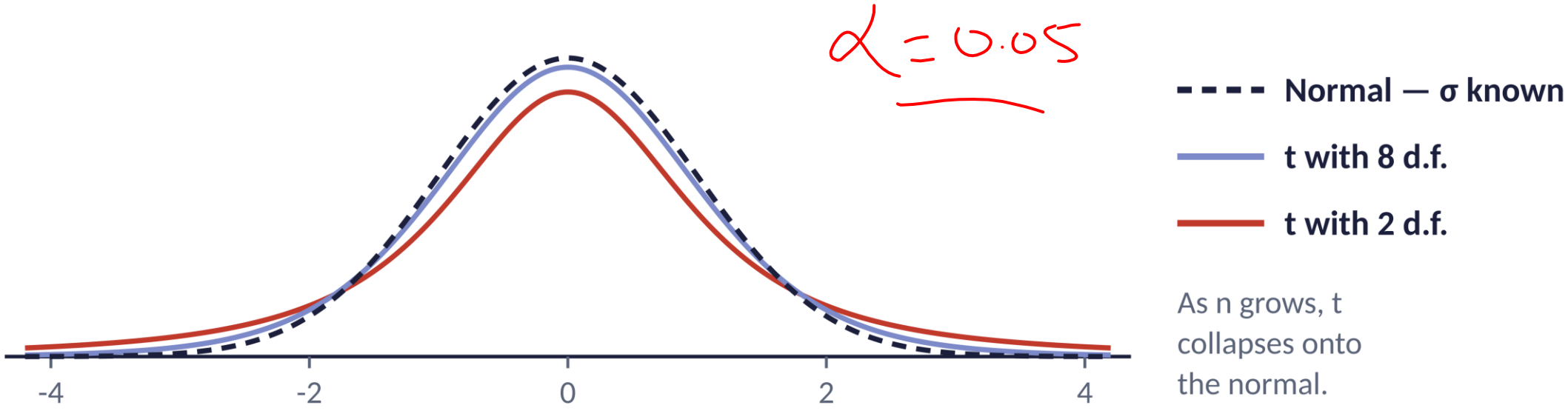
$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Degrees of freedom = n − 1

Same shape as the z-test, but with a changed denominator.

ONE PICTURE

s is a guess, so the tails get fatter



Fewer observations → fatter tails → you must travel further out to be surprised.

Two-sided at α = 0.05: the normal asks you to clear 1.96. With 8 degrees of freedom, t asks for 2.306.

ONE EXAMPLE

$\mu_0 = 4.0,\ n = 9,\ \bar{x} = 4.9,\ s^2 = 1.0$

Test at $\alpha = 0.05$, two-sided: $H_0: \mu = \mu_0$ against $H_1: \mu \neq \mu_0$

Work it out

Standard error = $s/\sqrt{n} = 1/\sqrt{9} = 1/3$

$t = (\bar{x} - \mu_0) / (s/\sqrt{n}) = 0.9 \div (1/3) = 2.7$

Degrees of freedom = 8, critical value = 2.306

THE IDEA THAT COMPRESSES EVERYTHING

Every test statistic has the same shape

(your estimate − the value H_0 claims) ÷ (the standard error of your estimate)

z-test, a mean	$\bar{x} - \mu_0$	*over*	$\sigma / \sqrt{n}$
z-test, a proportion	$p - p_0$	*over*	$\sqrt{(p_0(1-p_0) / n)}$
t-test, a mean	$\bar{x} - \mu_0$	*over*	$s / \sqrt{n}$

THE EXCEPTION THAT PROVES THE RULE

χ^2-test for a population variance

Use it when

- You are testing the spread itself, not the centre.
- The sample is assumed to come from a normal distribution.
- H_0: $\sigma^2 = \sigma_0^2$ · H_1: $\sigma^2 \neq \sigma_0^2$ · d.f. = $n - 1$

$$\chi^2 = \frac{(n-1)\,s^2}{\sigma_0^2}$$

Notice different

This is a ratio, not a difference over an error. A variance cannot be negative, so its distribution is skewed.

The two critical values are not symmetric.

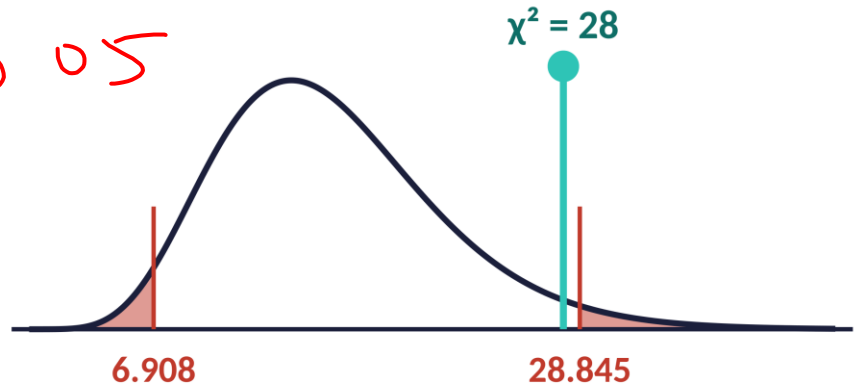
LAST EXAMPLE

Has Indian Railways become less reliable?

Last month, delays had a standard deviation of 2 hours. This month, 17 trains show s = √7. So $\chi^2 = 16 \times 7 / 4 = 28$, with d.f. = 16.

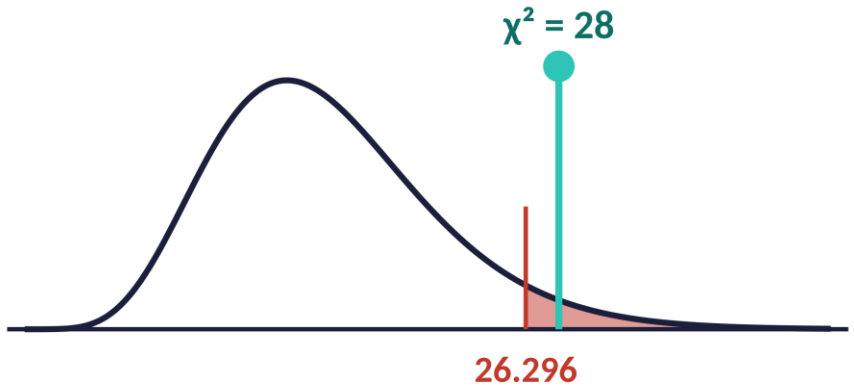
Two-sided — "has it changed?"

$\alpha = 0.05$



28 < 28.845 → do not reject

One-sided — "has it got worse?"



28 > 26.296 → reject

STUDY MATERIAL

Reference

The Art of Learning from Data: Chapter 9.2, 9.3 by Agresti, Franklin & Klingenberg

Slides